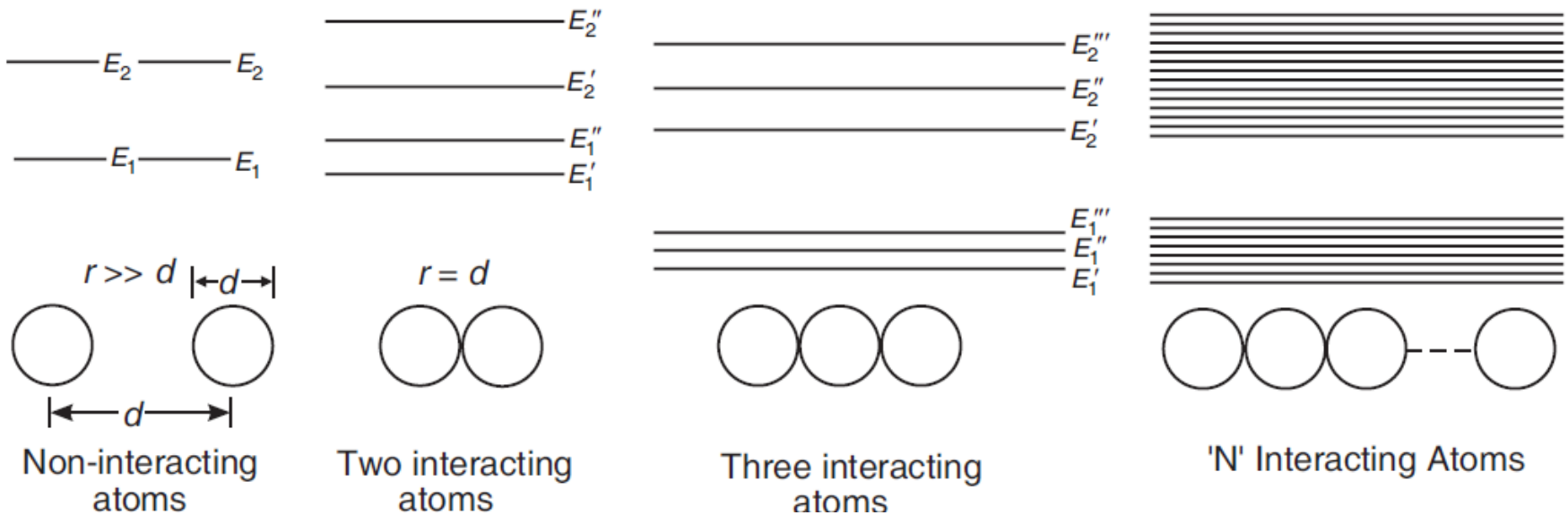
GITAM University

Subject-Physics

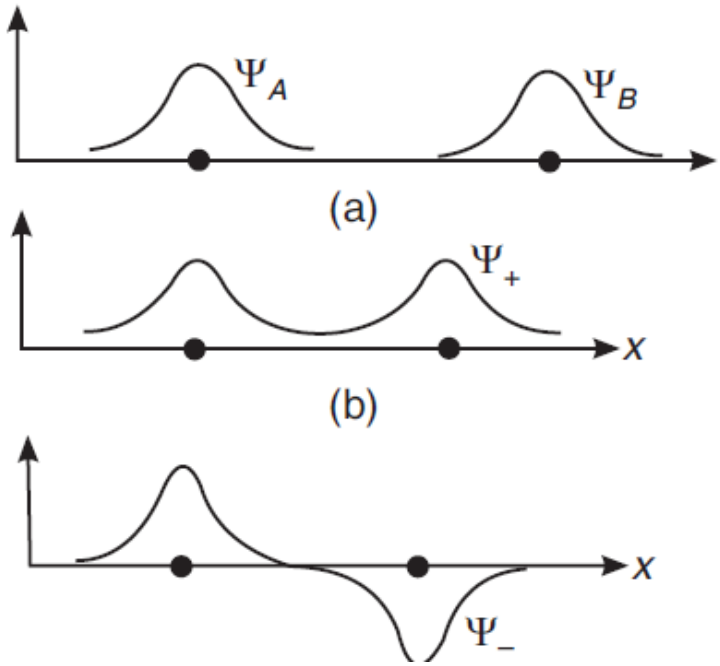
Topic: Semiconductors

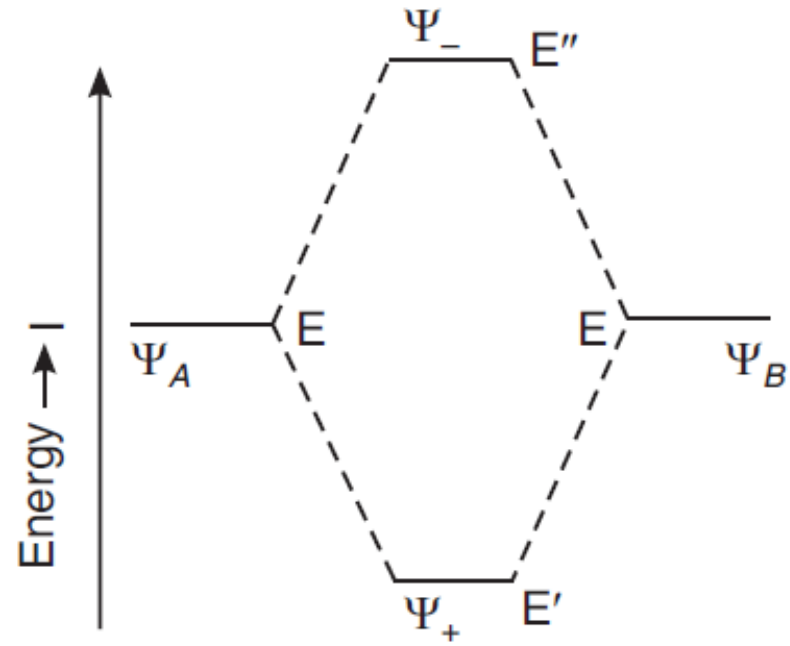
Sayan Chaudhuri



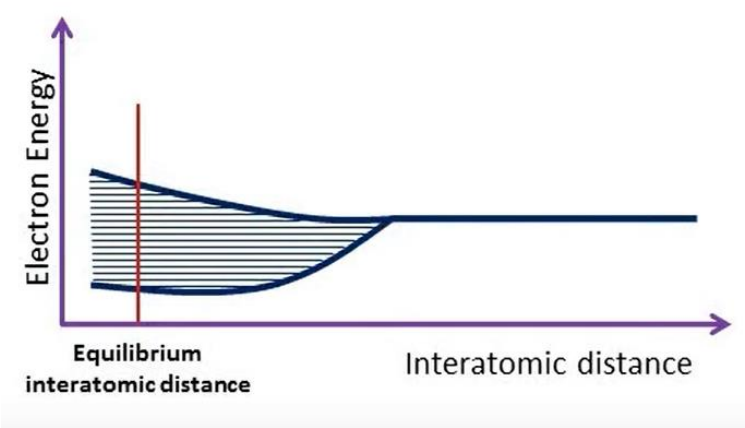
Pauli exclusion principle

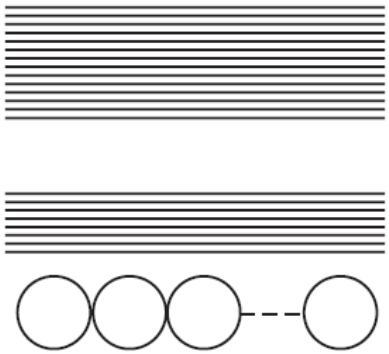
Two electrons can not have identical quantum states in interacting system.



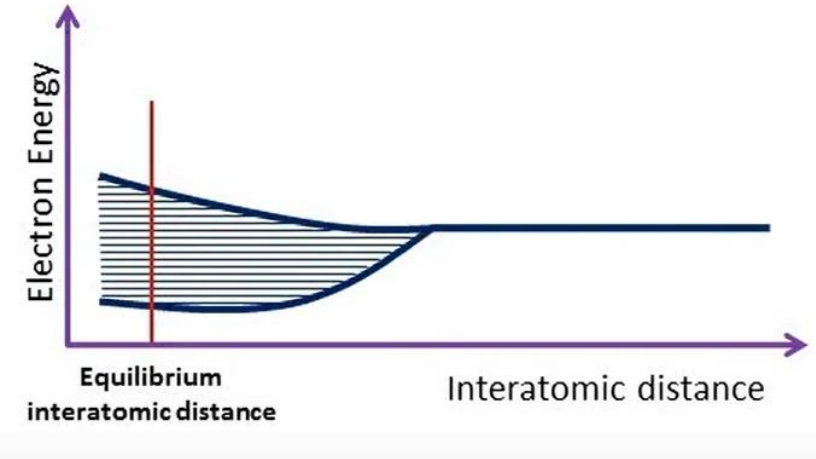


Atomic wave function of H_2.





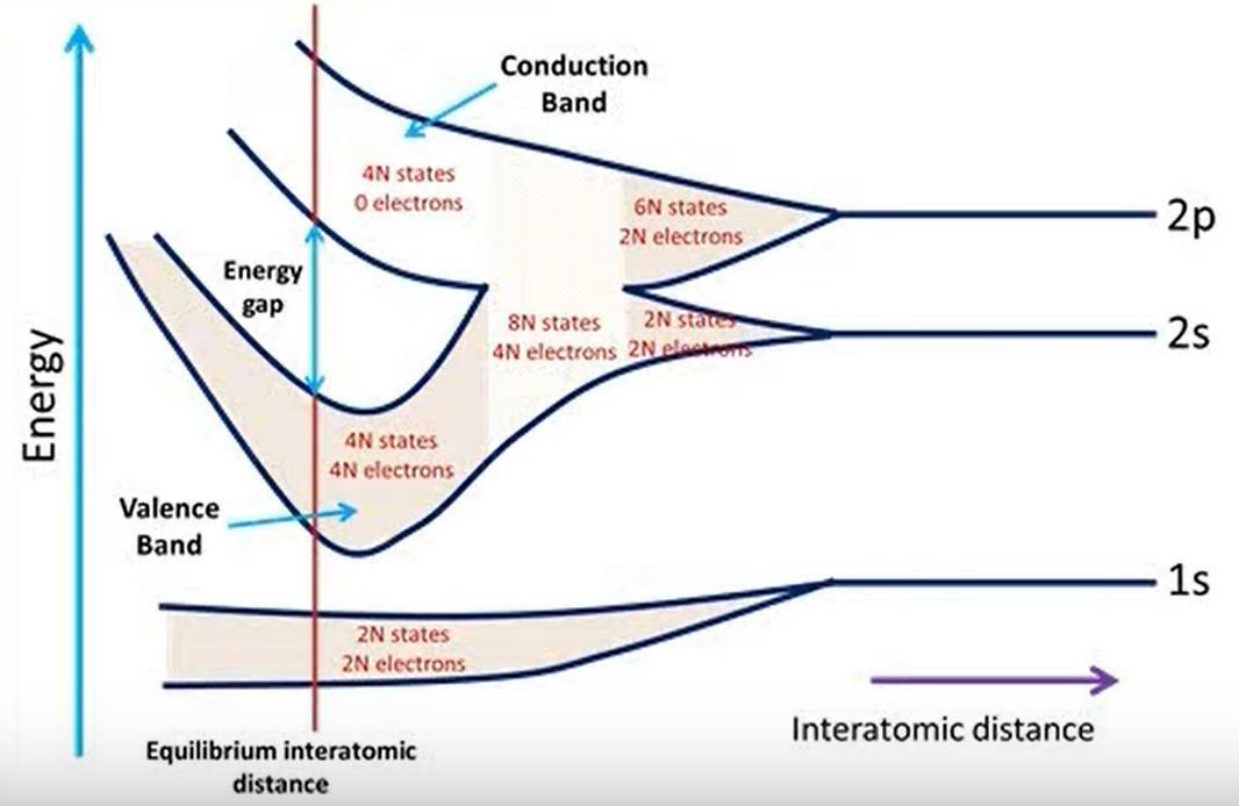
'N' Interacting Atoms

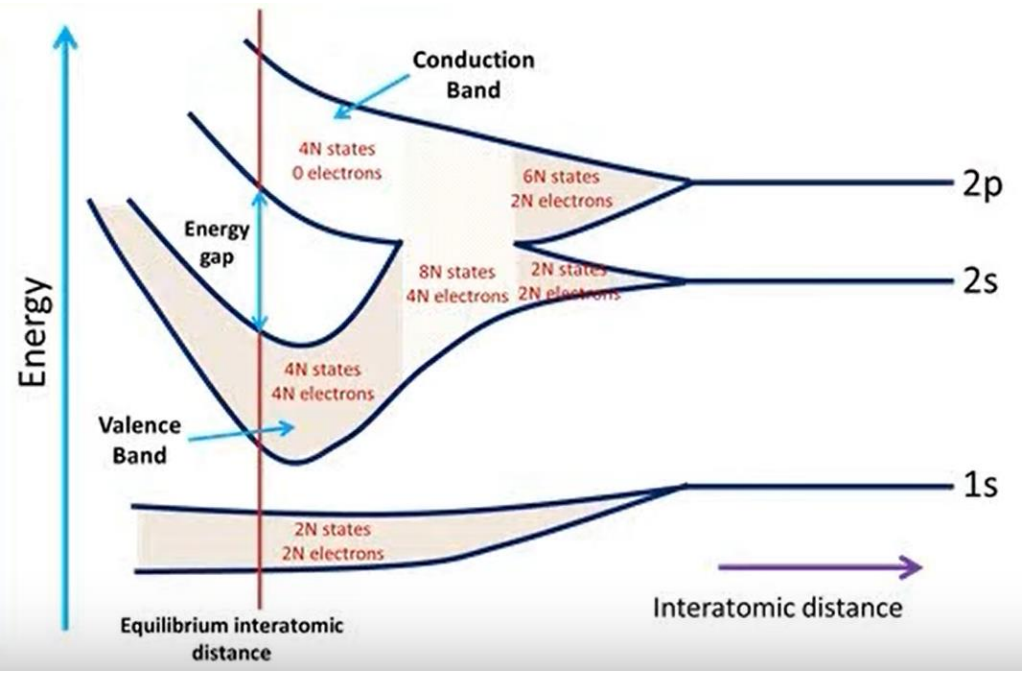


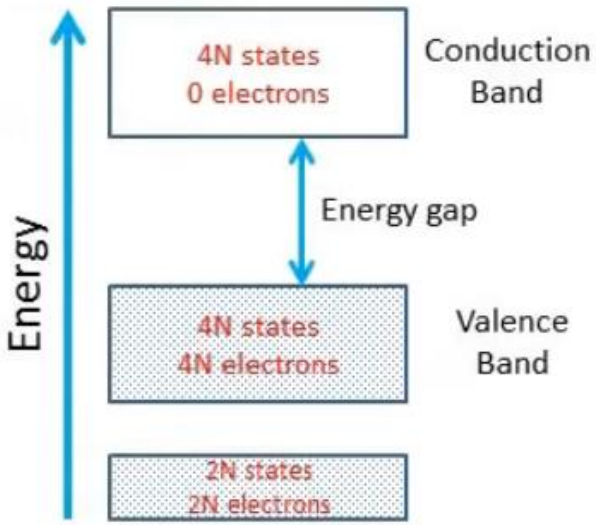
Pushing the atoms together, the initial quantized energy level will split into a band of discrete energy levels.

Diamond Band structure

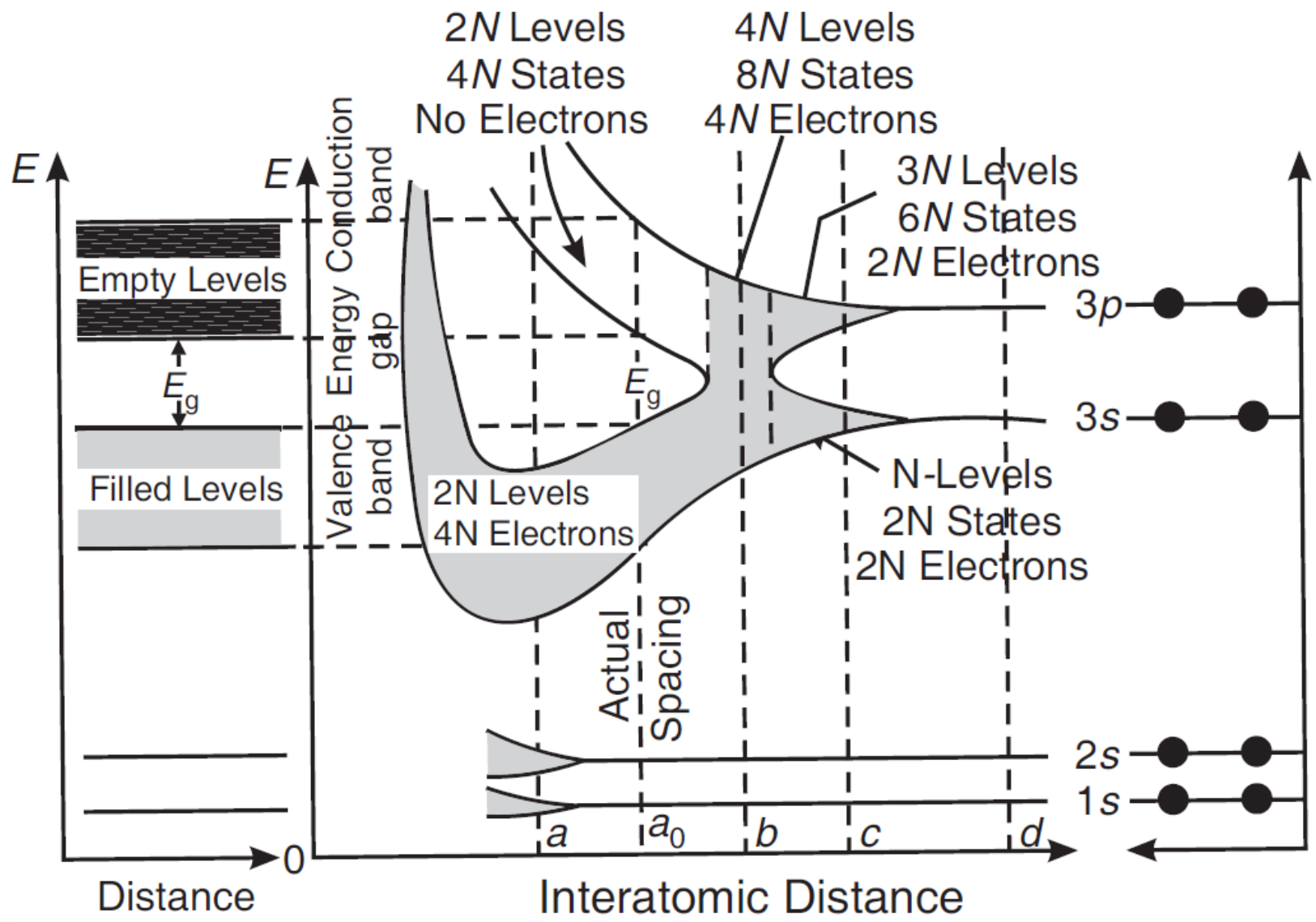
- Electronic structure individual carbon atom: $1s^2\ 2s^2\ 2p^2$
- Each atom has two 1s states, two 2s states, six 2p states and the higher states.
- Hence for N atoms, there will be 2N, 2N and 6N available states of type 1s, 2s, and 2p respectively.

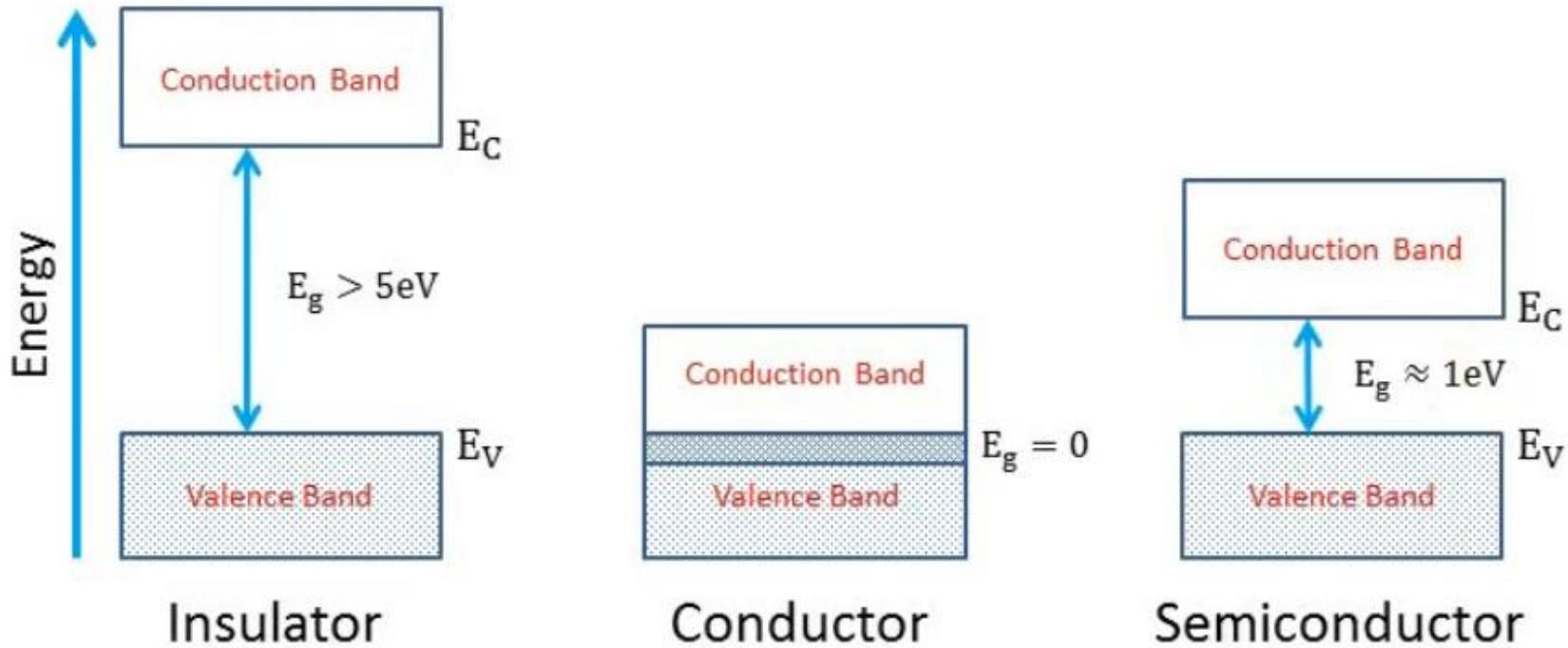



Conduction Band
4N states
0 electrons
6N states
2N electrons
2p
Energy gap
8N states
4N electrons
2N states
2N electrons
2s
Energy
4N states
4N electrons
Valence Band
1s
2N states
2N electrons
Equilibrium interatomic distance
Interatomic distance


Energy
4N states
0 electrons
Conduction Band
Energy gap
4N states
4N electrons
Valence Band
2N states
2N electrons

Silicone Band structure

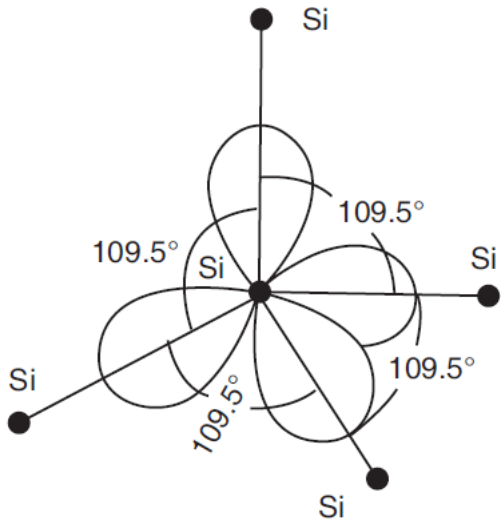


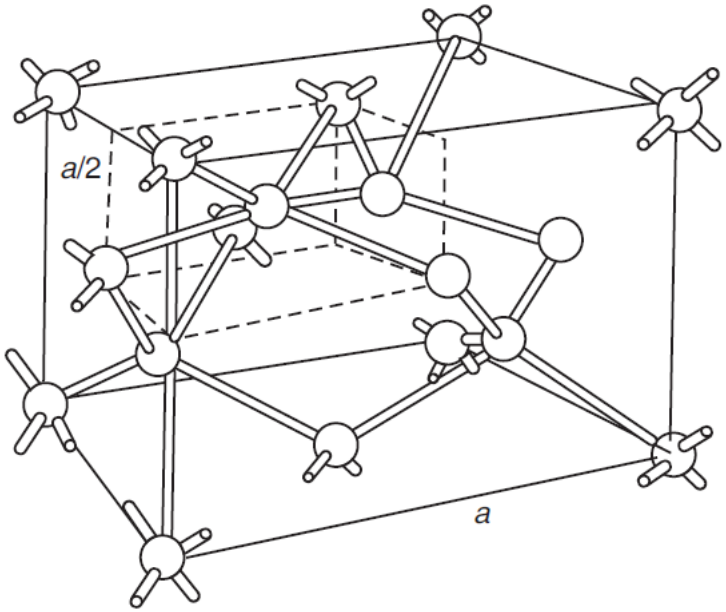

Energy
Conduction Band
E_C
$E_g > 5eV$
E_V
Valence Band
Insulator
Conduction Band
$E_g = 0$
Valence Band
Conductor
Conduction Band
E_C
$E_g \approx 1eV$
E_V
Valence Band
Semiconductor

Silicon (Si) and Germanium (Ge) are elemental semiconductors

- Si have 14 electrons
- 4 valance electrons

- Ge have 32 electrons
- 4 valance electrons

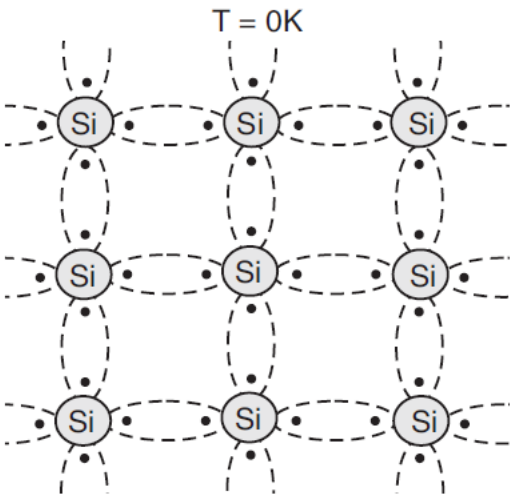




Si is preferred over Ge

- Silicon and Germanium are two most widely used semiconductors.
- Silicon is preferred over Germanium because of its superior thermal properties.
- The maximum operational temperature for Ge-devices is 80^0C while Si-devices may be used at up to 160^0C.
- Also, at room temperature, silicon crystal has almost no free electrons compared with Germanium crystal

INTRINSIC SEMICONDUCTOR



Impurity level less than 1 atom per 10^9 semiconductor atoms

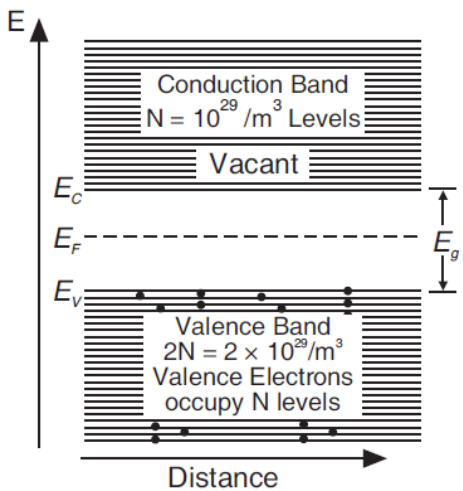
Concentration of Atom in a crystal

$$N = \frac{N_A \rho}{M}$$

For Si, N = $\frac{(6.02\times10^{26})\times2330}{28.09} = 5 \times 10^{28}$

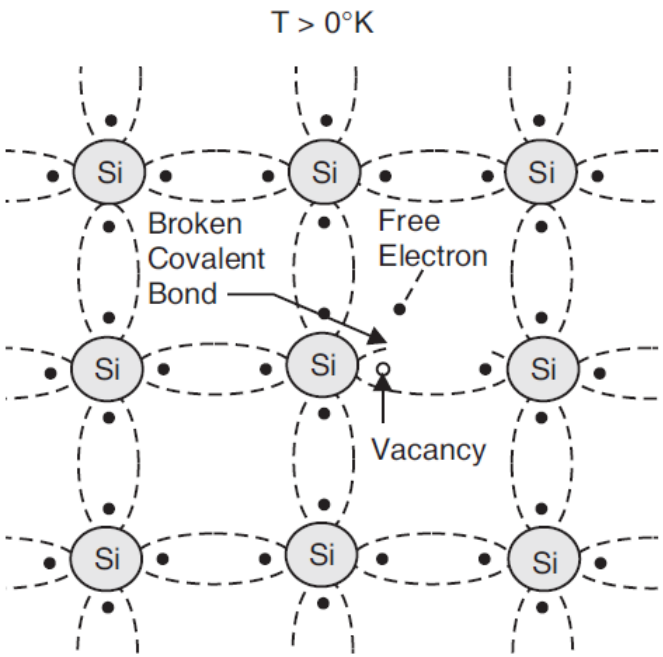
Number of valance electrons = 4N = 2×10^{29}

Neighboring atoms form covalent bonds by sharing electrons with each other

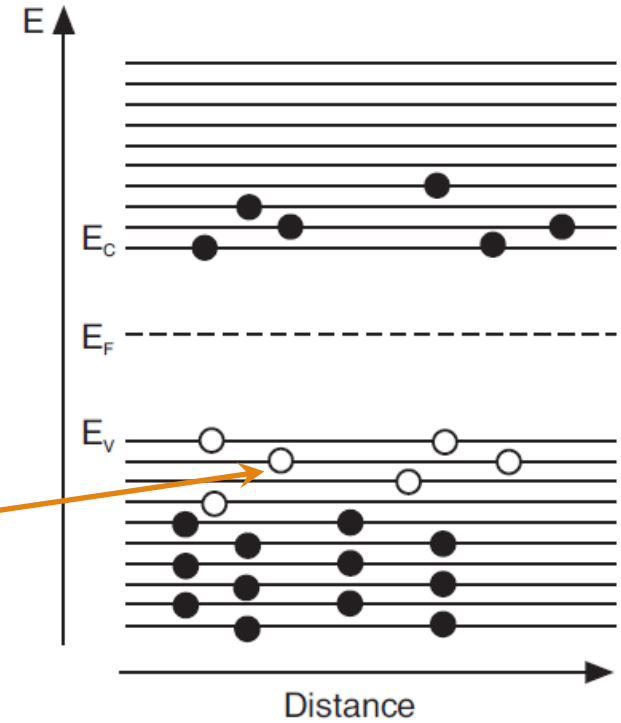


At 0K an Intrinsic Semiconductor Behaves as a Perfect Insulator

INTRINSIC SEMICONDUCTOR at finite Temperature



At 300 K 1.5× 10^{16} $/m^3$electrons can transfer to valance band.



Creation of Hole

An intrinsic semiconductor is a semiconductor crystal in which electrical conduction arises due to **thermally excited** electrons and holes

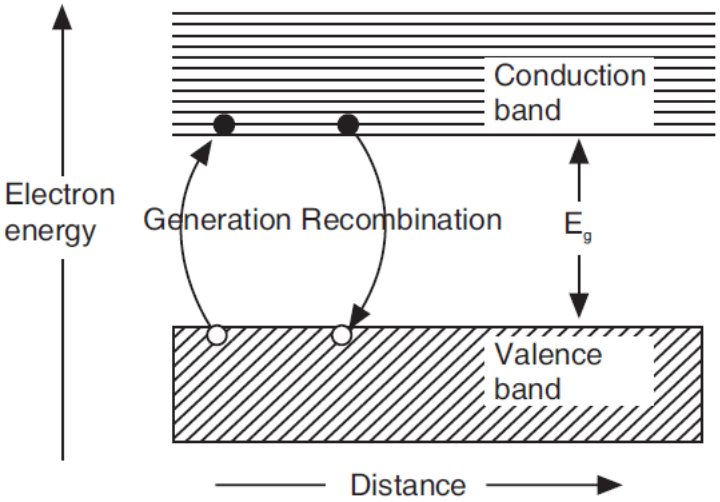
$$E = E_g + \frac{1}{2}mv^2$$

GENERATION AND RECOMBINATION

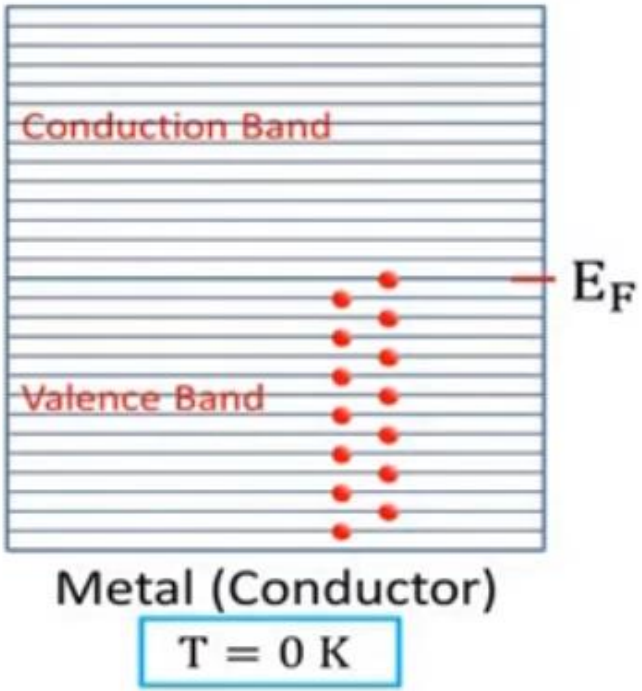
Generation

Covalent Bond + Thermal Energy ➔ Bond Breaking ➔ Electron +Hole

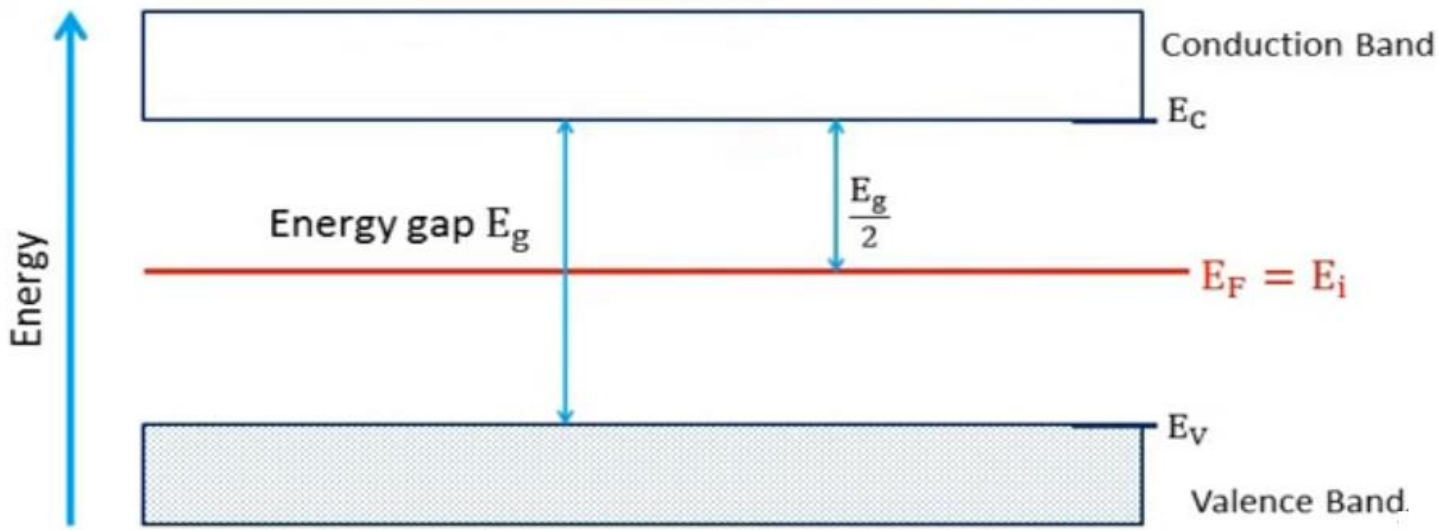
As temperature increases



- In metal, there is one partially filled band which is a result of conduction band overlapping with valence band.
- In this band lowest energy levels are filled first.
- The highest occupied energy level at absolute zero temperature (0K) is called as Fermi level.
- Energy corresponding to it is called as **Fermi energy denoted by E_F**



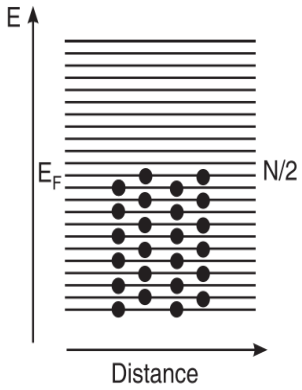
- In semiconductors, the **Fermi level** is defined as the reference level that gives the probability of occupancy of states in conduction band as well as unoccupied states in valence band.
- **Fermi level** in semiconductors may also be defined as energy level that corresponds to the centre of gravity of conduction band electrons and valence band holes weighted according to their energies.

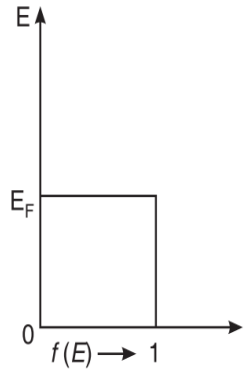


Fermi – Dirac Distribution:

The probability that an electron occupies an energy level E at thermal equilibrium

$$f\ (E) = \frac{1}{1 + \exp\left[\frac{E - E_F}{k_B T}\right]}$$





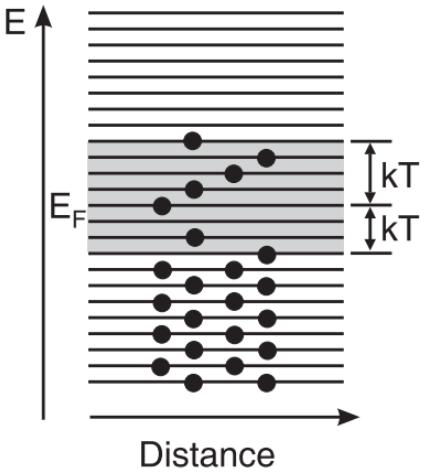
At T = 0,

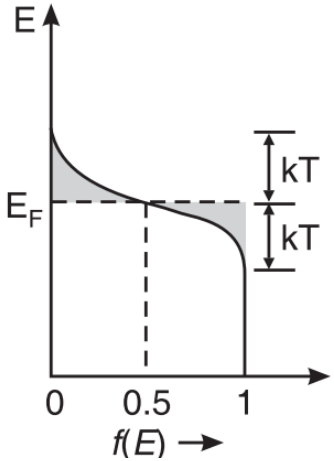
for $E<E_F$ f(E) = 0

for $E>E_F$ f(E) = 1

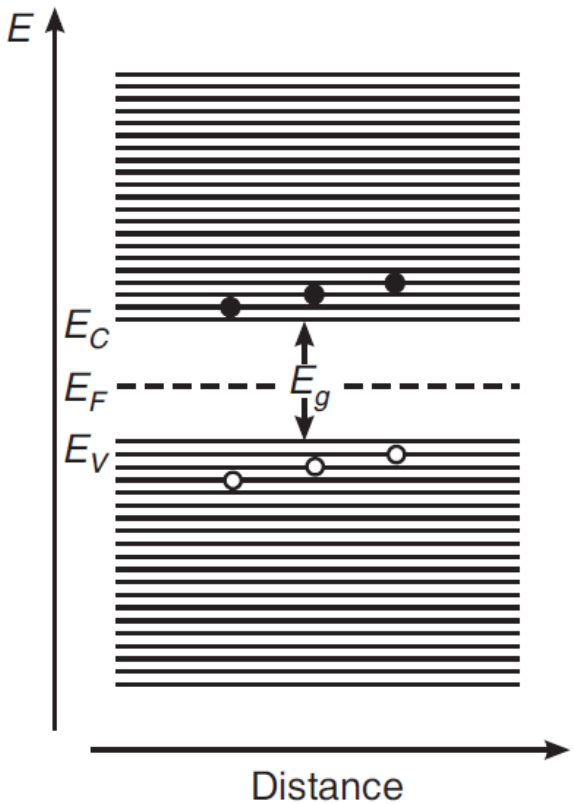
For $E = E_F$ f(E) = non – probable

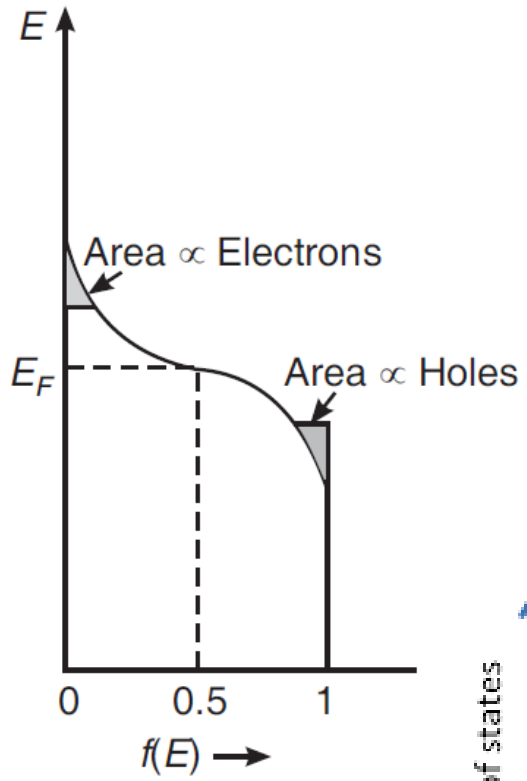
But ½ at any finite temperature

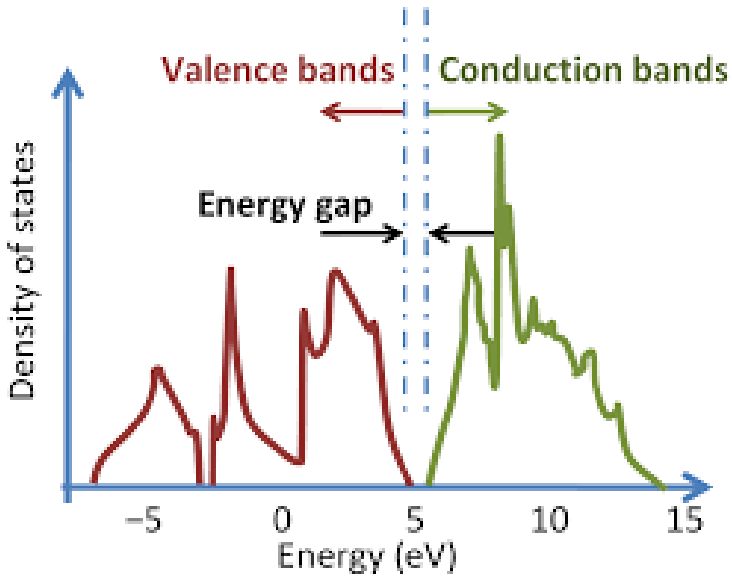




Density of states :



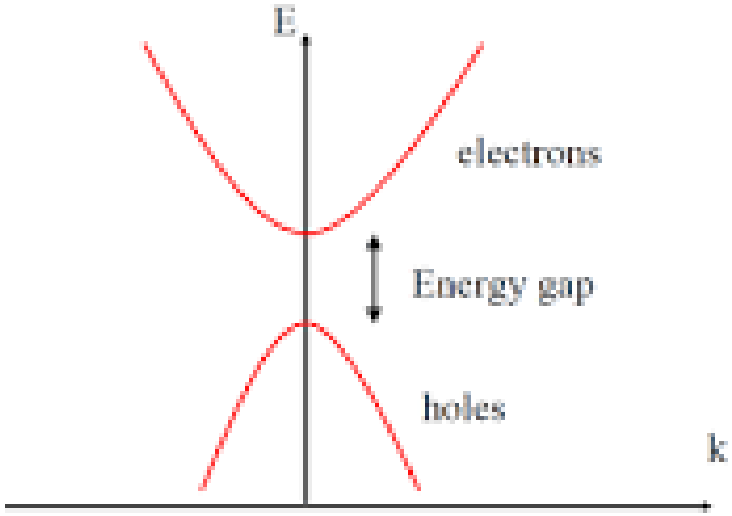




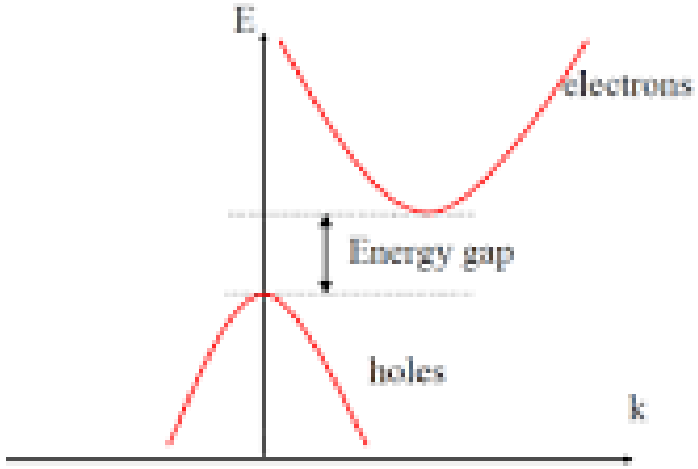
1. Calculate the probability of Non-occupancy for the energy level when it lies 0.01 eV above the Fermi level at 27^0C. K = $8.625X10^{-5}$ eV/K.

2. Fermi level of Ag is 5.5 eV. Calculate the energy for which the probability of occupancy at RT will be 0.9.

Direct and Indirect band SC



Direct bandgap semiconductors



Indirect bandgap semiconductors

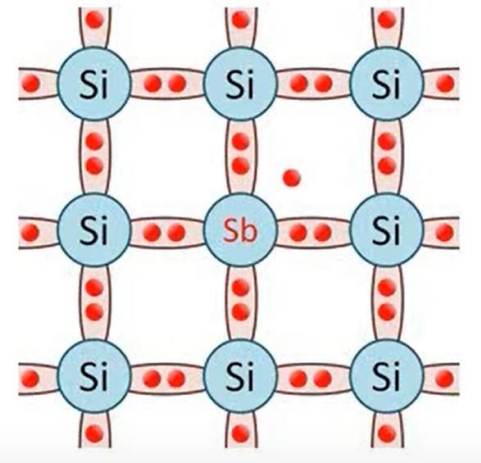
$$E = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}$$

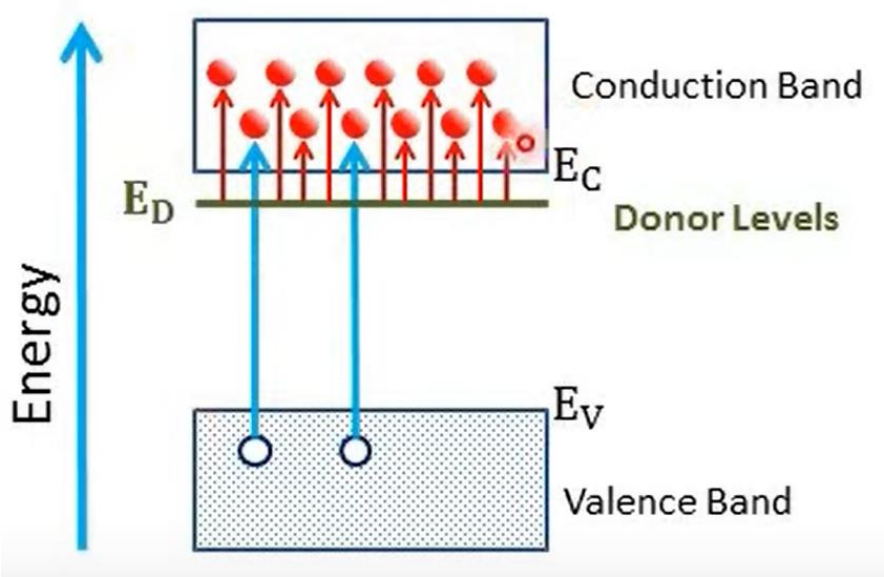
- Elemental SC' s (Si, Ge) are indirect band gap semiconductors
- GaAs is direct band gap SC

Direct band SC	Indirect band SC
Maximum of valence band and minimum of conduction band occur at same momentum values	Maximum of valence band and minimum of conduction band occur at two different momentum values.
Electron making a transition from valence band to conduction band need not undergo any change in its momentum.	In order to make a transition from maximum point in valence band to minimum point in conduction band, the electron requires energy for the change in momentum in addition to the energy gap E_g
The compound semiconductors such as GaAs, are direct gap semiconductors	All elemental semiconductors such as Si, Ge, are indirect gap semiconductors
These direct gap semiconductors are used in LED and Semiconductor Lasers.	Not useful for LEDs and Semiconductor Lasers

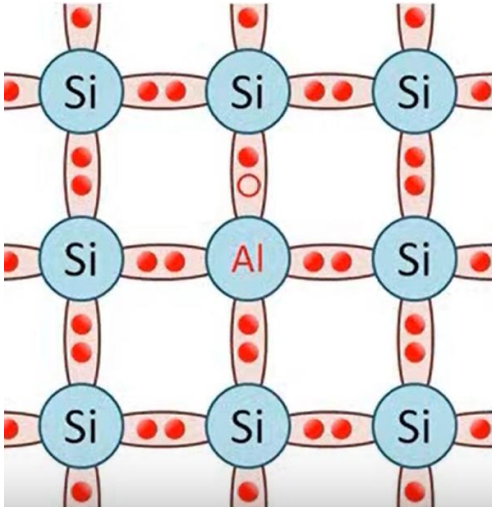
EXTRINSIC SEMICONDUCTOR:

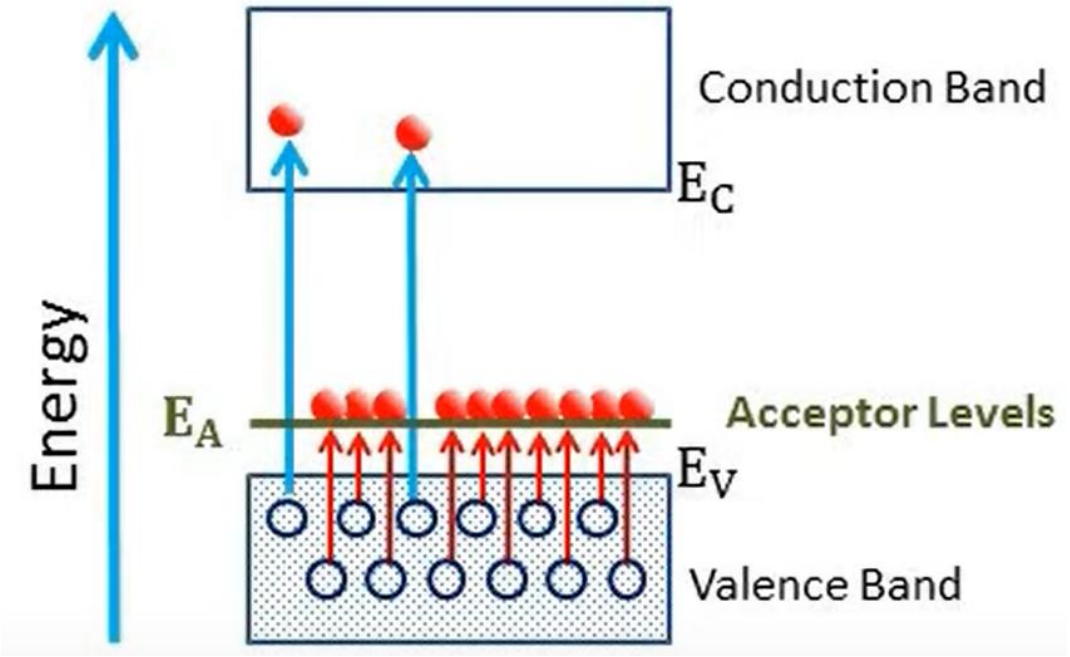
n- type semiconductor



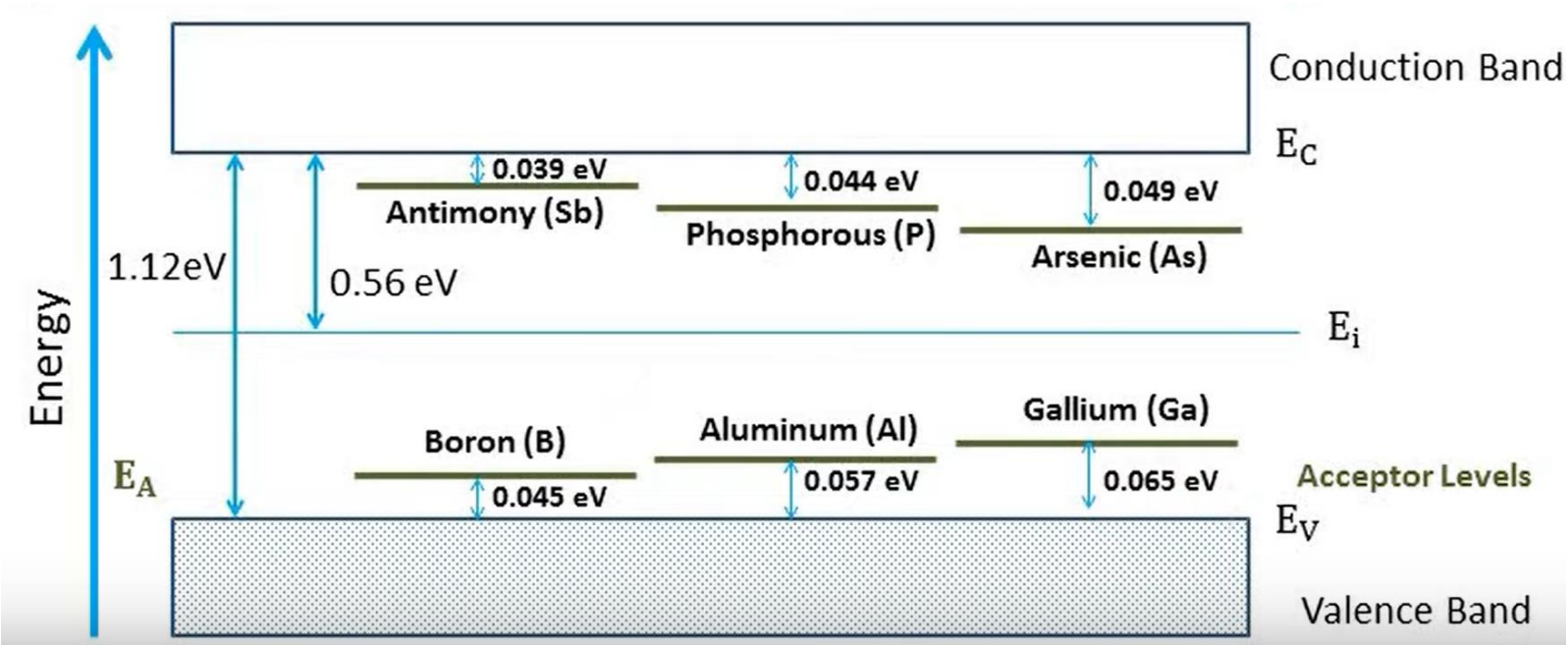


p- type semiconductor

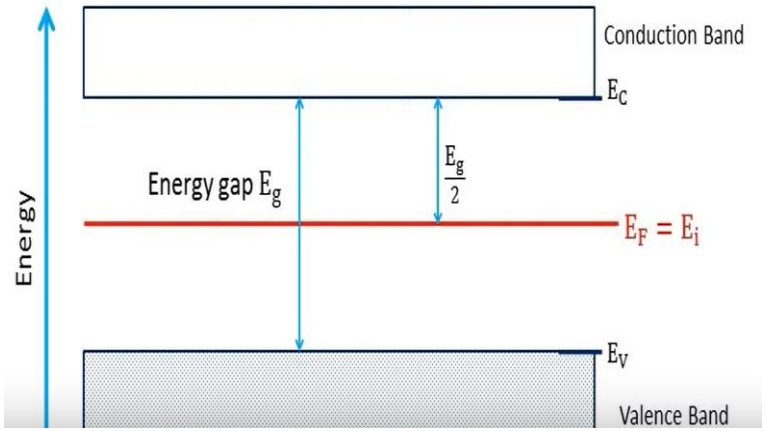




n-type	**p-type**
Phosphorous	Aluminium
Arsenic	Boron
Antimony	Gallium
	Indium

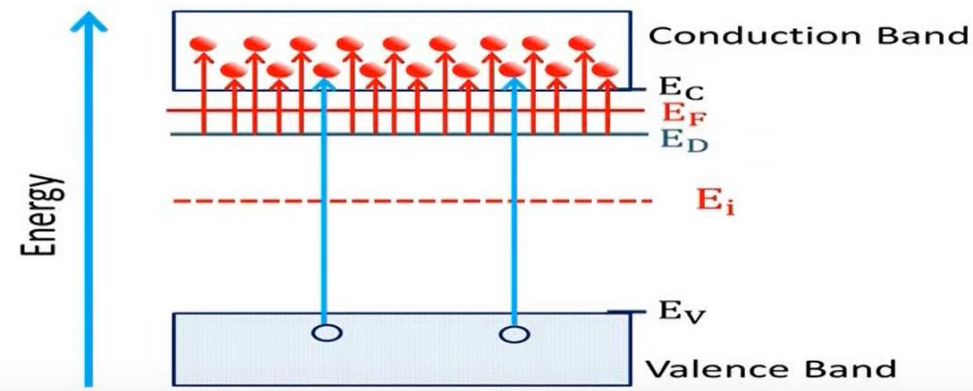


E_F is the energy level that corresponds to the center of gravity of conduction band electrons and valance band holes weighted according to their energies.

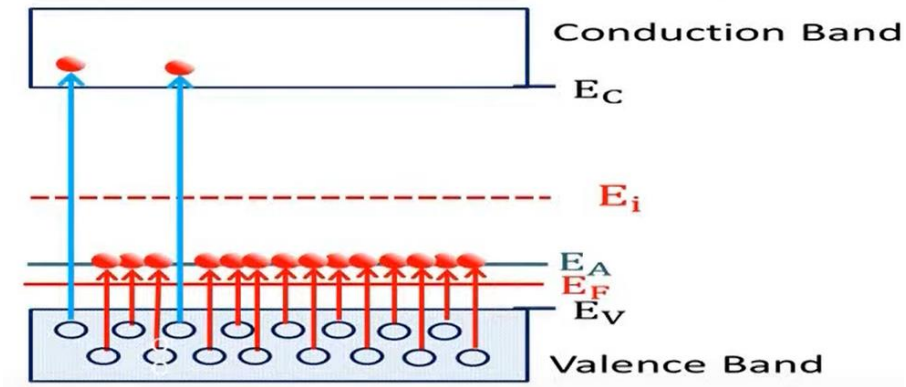


Intrinsic

n -type

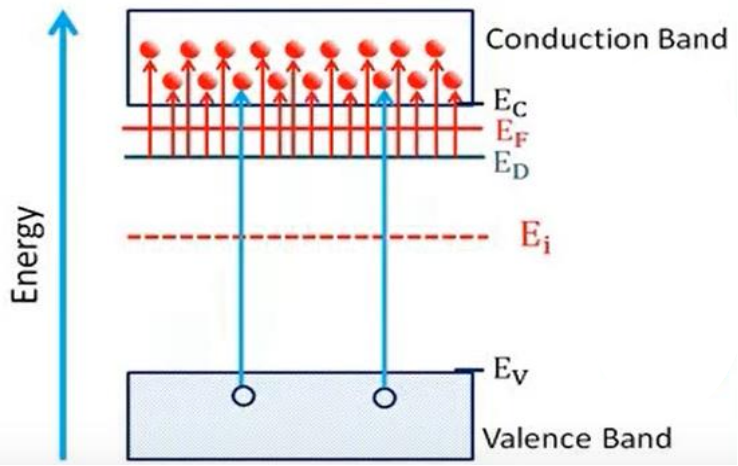


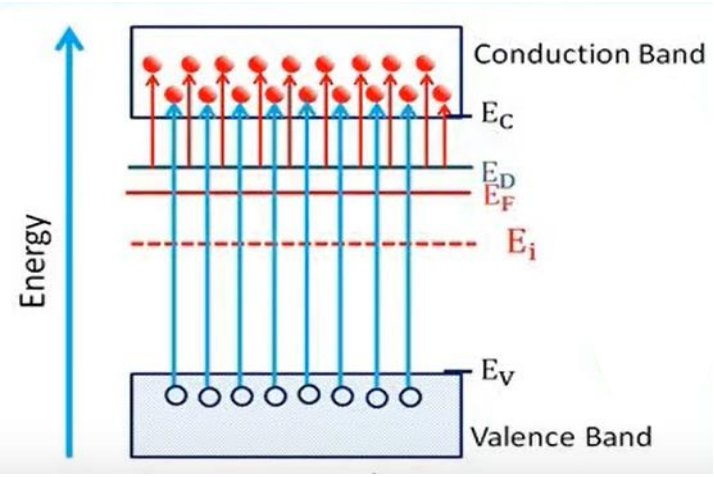
p -type

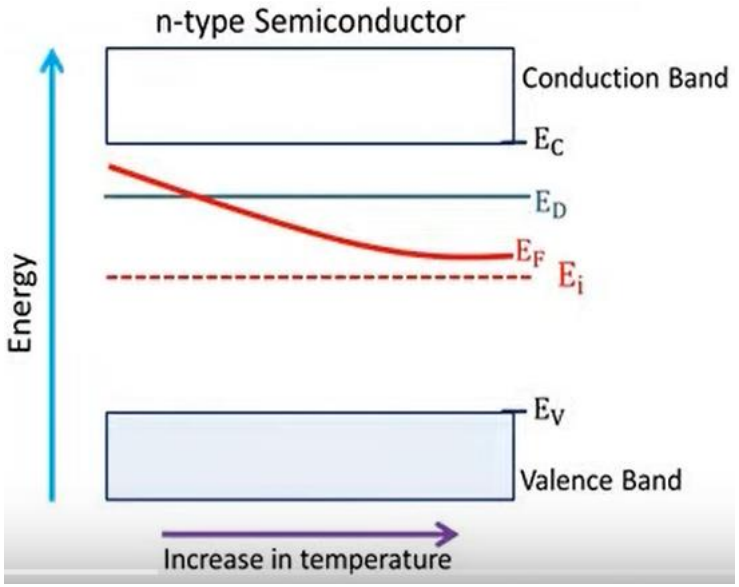


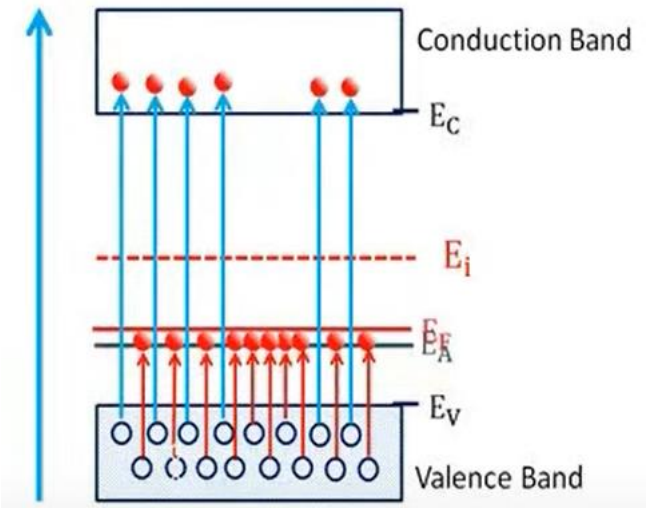
Temperature Effect on E_F

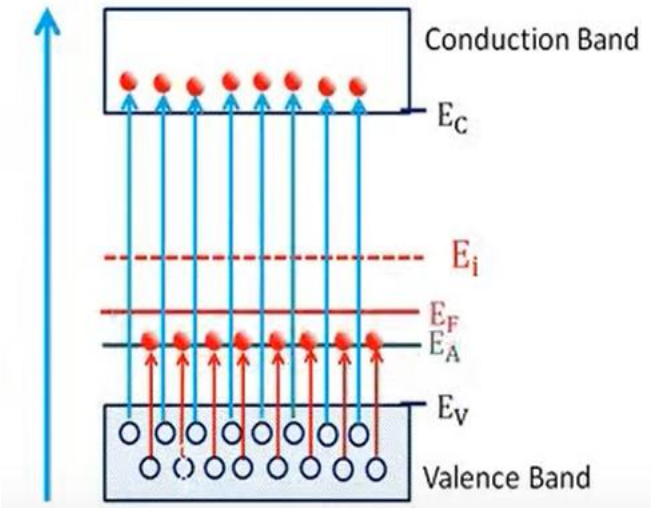
n – type SC

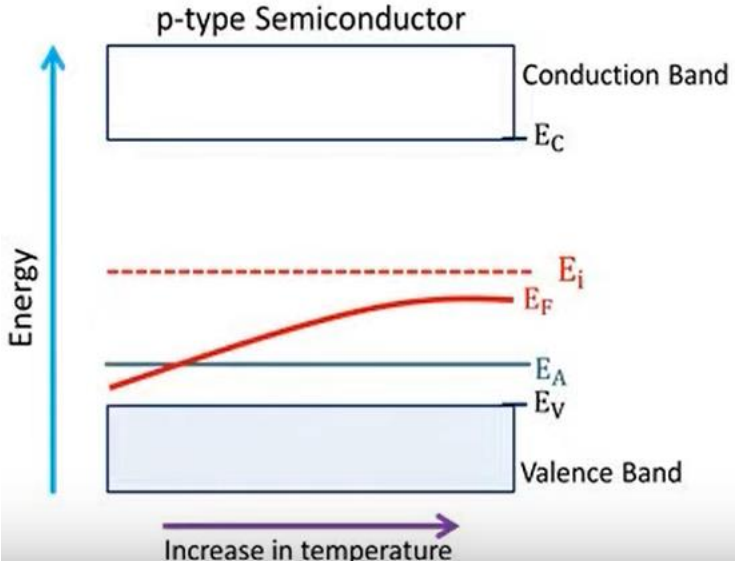




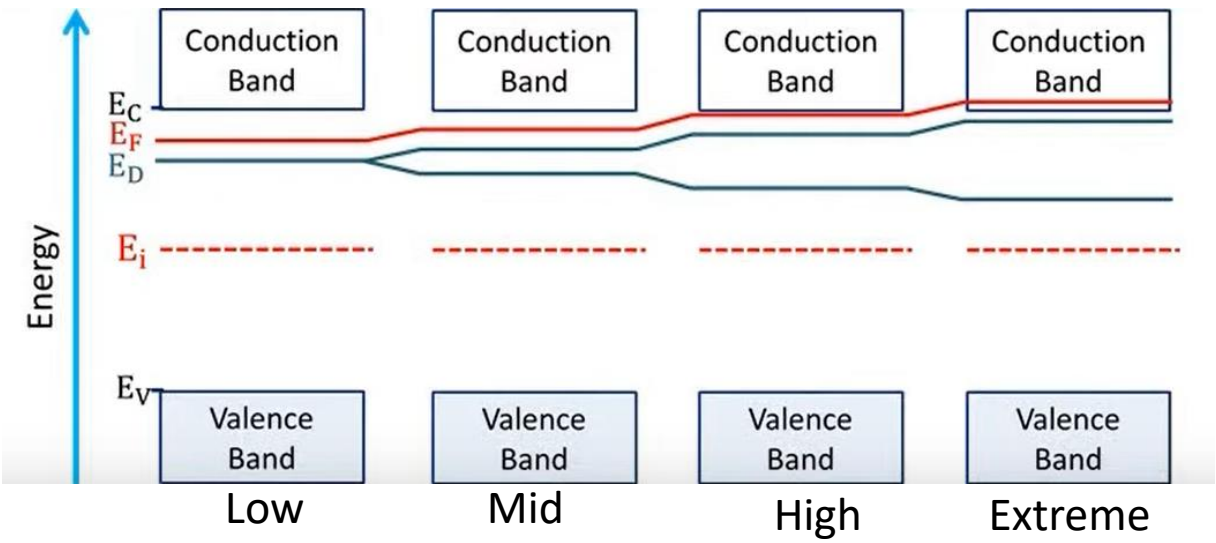



Conduction Band
E_C
E_i
E_F
E_A
E_V
Valence Band


Conduction Band
E_C
E_i
E_F
E_A
E_V
Valence Band


p-type Semiconductor
Conduction Band
E_C
Energy
E_i
E_F
E_A
E_V
Valence Band
Increase in temperature

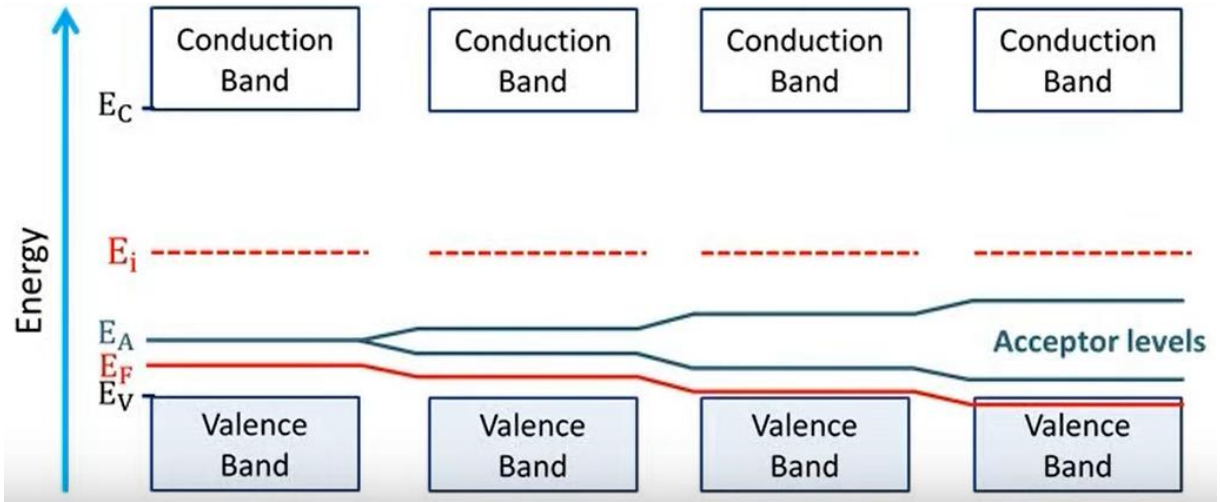
Doping Effect on E_F



Low Mid High Extreme

n - type

p - type



Low Mid High Extreme

Calculating the electron (or hole) concentration

Carrier Concentration,

$$n = \int_{E_c}^{\infty} f(E)n(E)dE$$

Electron concentration in the Conduction band,

$$n = N_C e^{\frac{-(E_C - E_F)}{kT}}$$

$N_C = \left(\frac{2\pi m_e^\star kT}{h^2}\right)^{\frac{3}{2}}$, effective density of state at Conduction band

Hole concentration in the Valance band,

$$p = N_V e^{\frac{-(E_F - E_V)}{kT}}$$

$N_V = \left(\frac{2\pi m_p^\star kT}{h^2}\right)^{\frac{3}{2}}$, effective density of state at Conduction band

For intrinsic SC:

$n_i = N_C e^{\frac{-(E_C - E_F)}{kT}}$ and $\mathrm{p_i} = N_V e^{\frac{-(E_F - E_V)}{kT}}$

We can deduce,

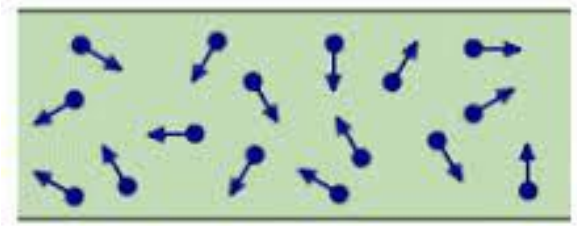
$$\boldsymbol{n.p = (n_i)^2}$$

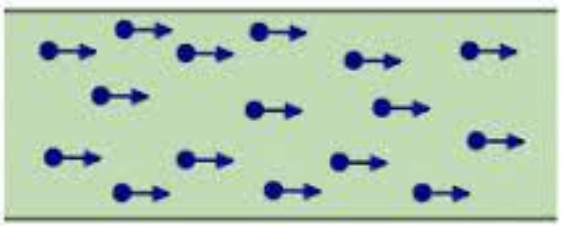
The position of E_F in intrinsic SC:

Taking $n_i = p_i$ and N_C = N_V at equilibrium

We will get,

$$E_F = \frac{E_C + E_V}{2}$$

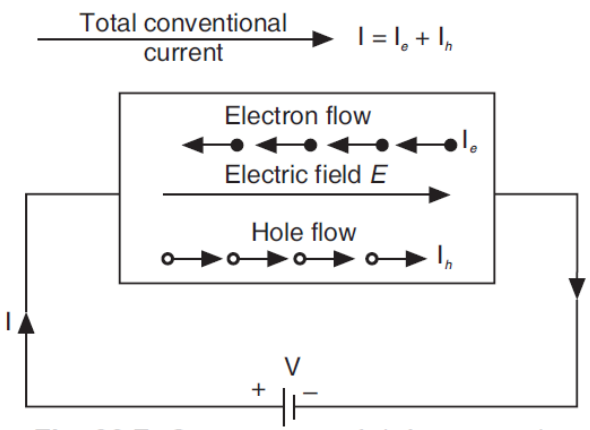




Average kinetic energy

$$\frac{1}{2}mv_{th}^2 = \frac{3}{2}k_B T - E_g$$

At thermal equilibrium total current due to random movement is 0



Under electric field

$$F = -e.E$$

Net motion of electron due to electric field is **Drift Motion**.

Drift Current Density:

$$J = \frac{Drift\ current, I}{Area, A}$$

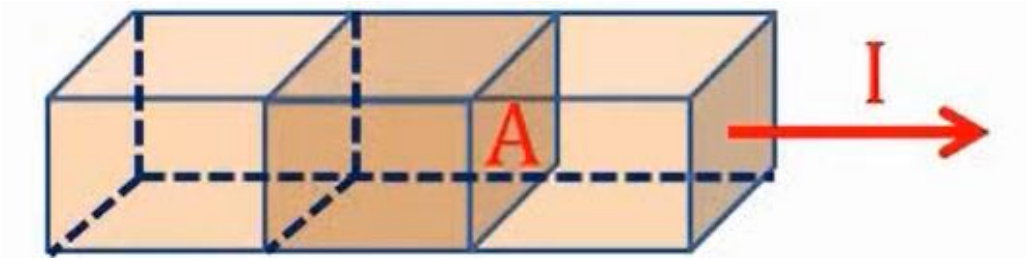
Electric Field:

$$\mathrm{E} = \frac{-dV}{dx} = \frac{V}{L}$$

Resistivity:

$$\rho = R.\frac{A}{L} = \frac{1}{\sigma}$$

$$J = \sigma E$$



Drift velocity : v : Net displacement of electron per unit time under influence of electric field

Mobility : $\mu = \frac{v}{E}$: Average drift velocity of electron per unit electric field

Volume swept by electron in unit time = $v.A$

Number of electron in the volume, = nvA

Current $\boldsymbol{I = e.nvA}$

$$\therefore \boldsymbol{J = \frac{I}{A} = nev}$$

$$\therefore \boldsymbol{\sigma = \frac{J}{E} = \frac{nev}{E} = ne\mu}$$

Conductivity in extrinsic Semiconductor

$$\sigma = \sigma_e + \sigma_h$$
$$\therefore \sigma = ne\mu_e + pe\mu_h$$

For intrinsic Semiconductor

$$n = p = n_i$$
$$\therefore \sigma = n_i(\mu_e + \mu_h)e$$

	Si	Ge
μ_e	1400	3900
μ_h	450	1900

$$\boldsymbol{\mu_e > \mu_h}$$

In n type

n >> p

$$\therefore \sigma = ne\mu_e$$

n is the donor concentration

In n type

p >> n

$$\therefore \sigma = ne\mu_p$$

p is the acceptor concentration

Problems

1. Calculate probability of non-occupancy for the energy level when it lies 0.01 eV higher than the E_F, at 27^0 C temperature.

2. E_F of Potassium is 2.1 eV. Find out the energies for which the probability of occupancy is 0.99 and 0.01 at RT.

$$f(E) = \frac{1}{1 + \exp\left[\frac{E - E_F}{k_{BT}}\right]}$$

3. What is the probability of an electron to be thermally excited to conduction band in intrinsic Si at RT, when band gap of Si is 1.12 eV.

$$k = 8.62 \times 10^{-5}$$

4. In an n type sc, E_F lies 0.4 eV below the CB. If the concentration of electron is doubled, what will be the position of the new E_F.

5. Calculate resistivity of Ge at RT where, density of carrier is 2.5×10^{19} /m^3. mobility of electron 0.39 m^2/V.S and hole 0.19 m^2/V.S.

$$\sigma = n_i(\mu_e + \mu_h)e$$

6. Calculate the number of donor atoms which must be added to an intrinsic semiconductor to obtain a resistivity of 10^{-6} ohm-cm. $\mu_e = 1000 \frac{cm^2}{V-Sec}$

The Fermi-Dirac distribution function is given by:

$$f(E) = \frac{1}{\left[1 + exp\left(\frac{E - E_F}{k_B T}\right)\right]}$$

1. Calculate probability of non-occupancy for the energy level when it lies 0.01 eV higher than the E_F, at 27^0 C temperature.

We want to find the probability of this energy level not being occupied, which is the complement of it being occupied: P(not occupied) = 1 - f(E)

Now, E = -0.01 eV (since it's below the Fermi level) and T = 300 K

f(E) = 1 / [1 + exp((-0.01 eV - 0.3 eV) / (8.617333262145 x 10^-5 eV/K * 300 K))] = 0.9526

P(not occupied) ≈ 1 - 0.9526 = 0.0474

2. In an n type sc, E_F lies 0.4 eV below the CB. If the concentration of electron is doubled, what will be the position of the new E_F.

Ans:

Fermi level lies 0.4eV below conduction band: $E_{F1}=E_C-0.4$

Concentration is doubled, $n_2=2n_1$

We know that,

$$n = N_C e^{-[(E_C-E_F)/kT]}$$

$$\frac{n}{N_C} = e^{[(E_F-E_C)/kT]}$$

$$ln(\frac{n}{N_C}) = \frac{E_F-E_C}{kT}$$

$$E_F - E_C = kTln(\frac{n}{N_C})$$

Initially,

$$E_{(F_1)} - E_C = kTln(\frac{n_1}{N_C}) ------(i)$$

After doubling the concentration,

$$E_{(F_2)} - E_C = kTln(\frac{n_2}{N_C}) ------(ii)$$

Subtracting (i) from (ii),

$$E_{(F_2)} - E_{(F_1)} = kT[ln(\frac{n_2}{N_C}) - ln(\frac{n_1}{N_C})]$$

$$E_{(F_2)} - E_{(F_1)} = -kTln(\frac{n_1}{n_2})$$

$$E_{(F_2)} - E_C - 0.4 = -kTln(1/2) \quad [n_2 = 2n_1]$$

Considering room temperature i.e T=300K

$$E_{(F_2)} = E_C - 0.4 - \frac{1.38\times10^{-23}}{1.6\times10^{-19}} \times 300ln(1/2)$$

$$E_{(F_2)} = E_C - 0.38eV$$

Fermi level is below conduction band by 0.38eV

30.8.1 Calculation of Electron Density

Let dn be the number of electrons whose energy lies in the energy interval E and $E + dE$ in the conduction band. Then,

$$dn = Z(E)f(E)dE \tag{30.20}$$

where $Z(E)\,dE$ is the density of states in the energy interval E and $E + dE$ and $f(E)$ probability that a state of energy is occupied by an electron.

The electron density in the conduction band is given by integrating the above equation between the limits E_C and ∞. E_C is the energy corresponding to the bottom edge of the conduction band and ∞ the energy corresponding to the top edge of the conduction band. As the probability of electrons occupying upper levels of conduction band $f(E)$ readily approaches zero for higher energies, the upper limit, namely the top of conduction band is taken as ∞. Thus,

$$n = \int_{E_C}^{\infty} Z(E)f(E)dE \tag{30.21}$$

The density of states in the conduction band is given by

$$Z(E)dE = \frac{4\pi}{h^3}\left(2m_e^*\right)^{3/2} E^{1/2} dE \quad \text{for } E > E_C. \tag{30.22}$$

The bottom edge of the conduction band E_C corresponds to the potential energy of an electron at rest. Therefore, $(E - E_C)$ will be the kinetic energy of the conduction electron at higher energy levels. Hence, equ. (30.22) is to be modified as follows.

$$Z(E)dE = \frac{4\pi}{h^3}\left(2m_e^*\right)^{3/2} \left(E - E_C\right)^{1/2} dE \tag{30.23}$$

The probability of an electron occupying an energy level is given by

$$f(E) = \frac{1}{1 + \exp\left[\left(E - E_F\right)/kT\right]}$$

When the number of particles is very small compared to the available energy levels, the probability of an energy state being occupied by more than one electron is very small. Such a situation is valid when $(E - E_F) >> 3kT$. Under this circumstance, the number of available states in the conduction band is far larger than the number of electrons in the band. Then, Fermi-Dirac function can be approximated to Boltzmann function,

$$f(E) = \exp[-(E - E_F)/kT]. \tag{30.24}$$

Using the equations (30.23) and (30.24) into (30.21), we obtain

$$\therefore \qquad n = \frac{4\pi}{h^3}\left(2m_e^*\right)^{3/2} \int_{E_C}^{\infty} \left(E - E_C\right)^{1/2} e^{-(E-E_F)/kT} dE \qquad (30.25)$$

or

$$n = \frac{4\pi}{h^3}\left(2m_e^*\right)^{3/2} e^{(E_F-E_C)/kT} \int_{E_C}^{\infty} \left(E - E_C\right)^{1/2} e^{-(E-E_C)/kT} dE \qquad (30.26)$$

The integral in eq. (30.26) is of the standard form which has a solution of the following form.

$$\int_0^{\infty} x^{1/2} e^{-ax} dx = \frac{\sqrt{\pi}}{2a\sqrt{a}}$$

where $a = 1/kT$ and $x = (E - E_C)$.

$$\therefore \qquad n = \frac{4\pi}{h^3}\left(2m_e^*\right)^{3/2} e^{(E_F-E_C)/kT} \left[\frac{\sqrt{\pi}}{2}(kT)^{3/2}\right] \qquad (30.27)$$

Rearranging the terms, we get

$$n = 2\left[\frac{2\pi m_e^* kT}{h^2}\right]^{3/2} e^{-(E_C - E_F)/kT} \tag{30.28}$$

The above equation is the expression for the **electron concentration** in the conduction band of an intrinsic semiconductor.

Designating

$$N_C = 2\left[\frac{2\pi m_e^* kT}{h^2}\right]^{3/2} \tag{30.29}$$

in the above equation, we obtain

$$n = N_C e^{-(E_C - E_F)/kT} \tag{30.30}$$

N_C is a temperature dependent material constant known as the **effective density of states** in the conduction band. In silicon at 300 K, $N_C = 2.8 \times 10^{25}/\text{m}^3$.